Worksheet 4E: Optimisation — Calculus Finds the Best Answer

Series 4: Calculus
AIML Foundations Mathematics
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What This Worksheet Is About
Calculus was invented, in large part, to answer the question: where is the highest? where is the lowest? Every machine learning algorithm, at its core, is doing optimisation — finding the weights that minimise the loss function. In this worksheet you will use derivatives to find maximum and minimum values, and at the end you will see exactly how this connects to gradient descent, the algorithm that trains neural networks.

Part A: Critical Points (10 problems)

A **critical point** of $f(x)$ is a value $x = c$ where $f'(c) = 0$ or $f'(c)$ is undefined. Critical points are candidates for local maxima, minima, or inflection points.

Find all critical points:

1. $f(x) = x^2 - 6x + 5$
2. $g(x) = 2x^3 - 9x^2 + 12x - 4$
3. $h(x) = x^4 - 8x^2 + 3$
4. $f(x) = x^3$ (hint: this critical point is neither a max nor a min — why?)
5. $f(x) = \dfrac{x^2 - 1}{x^2 + 1}$ (use the quotient rule)
6. $f(x) = e^x - ex$
7. $f(x) = x - \ln(x)$, $x > 0$
8. $f(x) = \sin x + \cos x$ on $[0, 2\pi]$
9. $f(x) = xe^{-x}$
10. $f(x) = \dfrac{\ln x}{x}$, $x > 0$

Part B: First and Second Derivative Tests (10 problems)

First Derivative Test: Check the sign of f' on either side of the critical point.
f' positive $\to$ negative: **local max**; f' negative $\to$ positive: **local min**; no sign change: **inflection point**.

Second Derivative Test: At c with $f'(c) = 0$:
$f''(c) > 0$: concave up $\Rightarrow$ **local min**; $f''(c) < 0$: concave down $\Rightarrow$ **local max**; $f''(c) = 0$: inconclusive.

Find and classify all critical points:

11. $f(x) = x^2 - 6x + 5$
12. $g(x) = 2x^3 - 9x^2 + 12x - 4$
13. $h(x) = x^4 - 8x^2 + 3$
14. $f(x) = xe^{-x}$
15. $f(x) = \dfrac{\ln x}{x}$, $x > 0$
16. $f'(3) = 0$ and $f''(3) = -7$. What can you conclude?
17. $g'(5) = 0$ and $g''(5) = 0$. Give an example showing the second derivative test is inconclusive.
18. Sketch a function with a local max at $x = -2$, local min at $x = 1$, inflection point at $x = -0.5$.
19. $f(x) = ax^2 + bx + c$, $a \neq 0$. Find the one critical point and show it is a min when $a > 0$, max when $a < 0$.
20. For $f(x) = x^n$ (n a positive integer): find and classify the critical point. What pattern do you see for even vs odd n?

Part C: Optimisation Word Problems (12 problems)

Steps: (1) Write the quantity to optimise. (2) Use constraints to reduce to one variable. (3) Find and classify critical points. (4) Check boundaries.

21. **Fencing.** 120 m of fencing; one side against a wall (no fencing needed). (a) Let $x =$ width. Write area $A(x)$. (b) Find dimensions maximising area. (c) Maximum area?
22. **Box.** Square base, no lid, volume 32 m^3. Base costs $2/\mathrm{m}^2$, sides $1/\mathrm{m}^2$. (a) Constraint: $x^2h = 32$. (b) Write cost $C(x)$. (c) Minimise cost.
23. **Can.** Volume 500 cm^3. Surface area $S = 2\pi r^2 + 2\pi rh$. (a) Eliminate h. (b) Find r and h minimising S. (c) What is the ratio h/r at the optimum?
24. **Profit.** Revenue $R(x) = 50x - 0.5x^2$, cost $C(x) = 10x + 200$. (a) Profit $P(x)$? (b) Profit-maximising quantity? (c) Maximum profit?
25. **Coastline problem.** A ship is 4 km from the coast; its destination is 9 km along the coast. Speed at sea: 20 km/h; on land: 30 km/h. Where should it land to minimise travel time?
26. **Window.** Rectangle topped by semicircle; perimeter = 10 m. Maximise area.
27. **Marginal cost.** $C(x) = 0.04x^3 - 0.9x^2 + 10x + 200$, $0 \leq x \leq 20$. (a) Marginal cost $C'(x)$? (b) Minimum of marginal cost? (c) Minimum of average cost $\bar{C}(x) = C(x)/x$?

28. **MSE.** Linear model $\hat{y} = mx$. Data: $(1, 2), (2, 3), (3, 5)$. $L(m) = \frac{1}{3}[(m-2)^2 + (2m-3)^2 + (3m-5)^2]$. (a) Expand. (b) $dL/dm = 0$. (c) Best slope m?
29. **Cross-entropy.** $L(p) = -\ln(p)$ (for $y = 1$). Find p minimising L.
30. **Snell's Law.** Light travels from medium 1 (speed v_1) to medium 2 (speed v_2). Minimising travel time gives Snell's Law: $\sin\theta_1/v_1 = \sin\theta_2/v_2$. Set up the time function and differentiate.
31. **Regularisation.** Add λm^2 to the MSE from problem 28. Find the new minimising m. How does increasing λ affect m?
32. **Rectangle in a circle.** Inscribed in a circle of radius 5. Maximise area. (Constraint: $x^2 + y^2 = 25$.)

Part D: Absolute Extrema on Closed Intervals (6 problems)

On $[a, b]$: evaluate f at all critical points in (a, b) and at both endpoints. The largest is the absolute max; the smallest is the absolute min.

33. Find absolute max and min of $f(x) = x^3 - 3x + 2$ on $[-2, 2]$.
34. Find absolute max and min of $g(x) = x - 4/x$ on $[1, 4]$.
35. Find absolute max and min of $h(x) = 2\sin x + \cos 2x$ on $[0, \pi]$.
36. Temperature $T(t) = -t^2 + 6t + 10$ on $[0, 8]$. (a) When is temperature highest? (b) Highest and lowest temperatures?
37. Prove algebraically that among all rectangles with fixed perimeter P, the square has maximum area.
38. Why doesn't the Extreme Value Theorem guarantee a max/min for $f(x) = 1/x$ on $(0, 1)$?

Part E: Gradient Descent (8 problems)

$$\boxed{x_{n+1} = x_n - \alpha \cdot f'(x_n)}$$

$\alpha > 0$ is the **learning rate**. Gradient descent descends the slope of f toward a minimum.

39. $f(x) = x^2$, $f'(x) = 2x$. Start $x_0 = 4$, $\alpha = 0.1$. Compute x_1, x_2, x_3, x_4. What is it approaching?
40. Repeat with $\alpha = 0.5$. What happens?
41. Repeat with $\alpha = 1.1$. What happens? Why is a large learning rate dangerous?
42. $f(x) = (x-3)^2 + 1$. Start $x_0 = 0$, $\alpha = 0.2$. Perform 5 steps. Convergence value?
43. Starting from $m_0 = 0$ with $\alpha = 0.1$, perform 3 gradient descent steps on $L(m)$ from problem 28. Are you approaching the true minimum?

44. Learning rate dilemma: (a) α too small — what is the problem? (b) α too large — what is the problem? (c) Sketch the loss over iterations for all three cases.

45. **SGD.** Why does using a random mini-batch approximation of the gradient still work for training?

46. Reflection: (a) Gradient descent uses $f'(x) \approx 0$ at a minimum — without solving for it. How? (b) Why does gradient descent always converge for a convex (concave-up) function? (c) How would gradient descent look in matrix form for a network with many weights?